

Electricity and magnetism

Have you ever wondered how an electric fan works? What is a short circuit and how can it be avoided? How does lightning occur on the sky? Why do magnets attract some materials? How does a magnetic compass work?

The answers above questions, lie in our understanding of the concepts of **electricity** and **magnetism**.

In this chapter, we will learn about:

- Electric current and its flow
- Electrical circuits and its components
- Fuses and its uses
- Static electricity (Lightning as an example)
- Charges and their properties
- Magnetic lines of force and magnetic field
- Electromagnets
- The Earth's magnetism
- Magnetic compass

All the students will be able to:

- ✓ Explain how an electric current flows through a circuit.
- ✓ Explain the working of a fuse and give details as to why it is used in an electric circuit.
- ✓ Describe how lightning occurs.
- ✓ Understand how static electric charges can be produced in some common materials.
- ✓ Investigate the magnetic field of a bar magnet.
- ✓ Construct an electromagnet and describe its working.
- ✓ Identify electromagnetic devices of everyday use.
- ✓ Explain the association between magnetism and electricity in various electromagnetic devices.
- ✓ Describe the magnetic field of Earth, and connect this concept with the use of a magnetic compass.
- ✓ Construct a magnetic compass and describe its working.



Figure 8.1: A magnet attracting nails



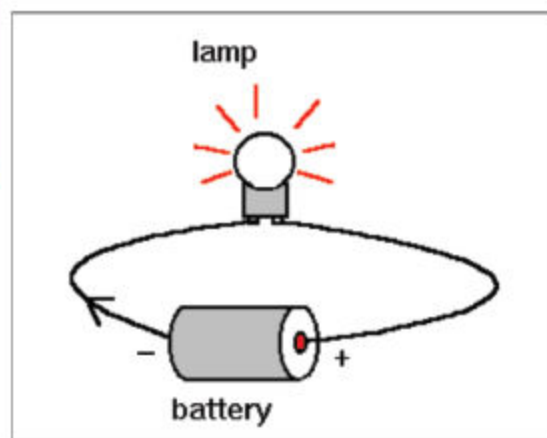
Figure 8.2: A electricity pole

Electrical circuit, current and its flow

Activity 1: Identifying an electric circuit

Explain
how an electric
current flows
through a
circuit.

Look at the picture and explain what is happening.



Why is the light bulb glowing?

What is the source of energy for the bulb?

What does the black line represent?

What does the arrow represent?

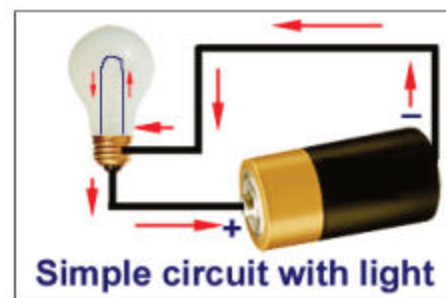


Figure 8.3 An electric circuit

As you have studied earlier, the flow of negative electric charges through an electric path (circuit) is called an electric current. An electric circuit is a loop or path along which a current flows. It mainly consists of:

- A source of electrical energy; **battery**
- A device which converts the electrical energy; **bulb**
- A pathway to aid the flow of electric current; **wires/conductors**

An electrical circuit is a path that allows electric charges to flow and perform a task, such as to light a bulb. Electricity flows from the source to the device (bulb) and then back to the original source. The device converts electrical energy into other forms of energy such as heat, light and other forms.

Activity 2: Use the symbols to draw an electric circuit

From the introduction, we believe that you have understand the primary idea of components of the electrical circuit. Following are some of the useful components which are used in circuits.

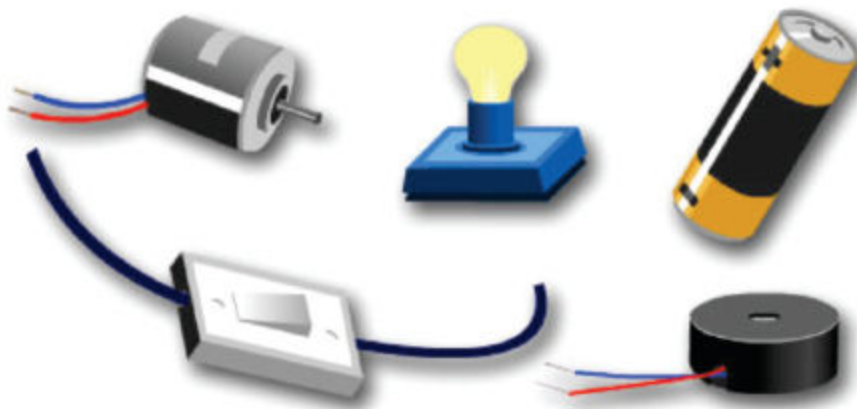
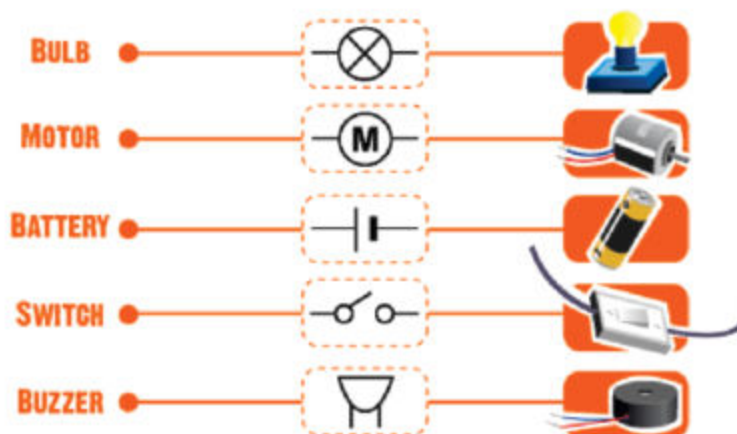


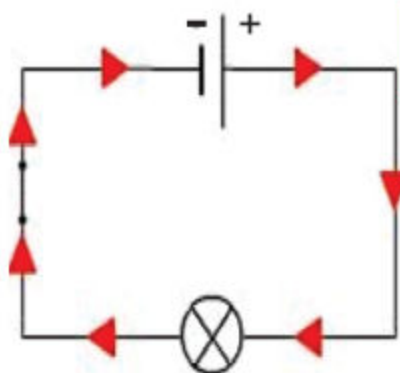
Figure 8.4: Components of an electric circuit

But we don't draw the circuits with the colorful pictures. For drawing a proper circuit diagram, we use symbols:



Now with the help of these symbols try to draw a circuit diagram.

A circuit diagram is provided here to help you do the activity.



Activity 3: With the help of the given symbols, draw three circuit diagrams,

- Draw a circuit diagram showing a cell, a switch and a bulb.
- Draw a circuit diagram showing a cell, a switch and a motor.
- Draw a circuit diagram showing a cell, a switch and a buzzer.

A fuse and it's uses:

📘 Explain the working of a fuse and give details as to why it is used in an electric circuit.

A **fuse** is a safety device, used in a circuit to protect the appliances and equipment and its wiring from damage, which can be caused by the excessive current flow.

Inside the fuse, there is a wire; when too much current flows through it, the wire melts and stops the flow of the current. The thicker the wire, the more current is needed to melt it.



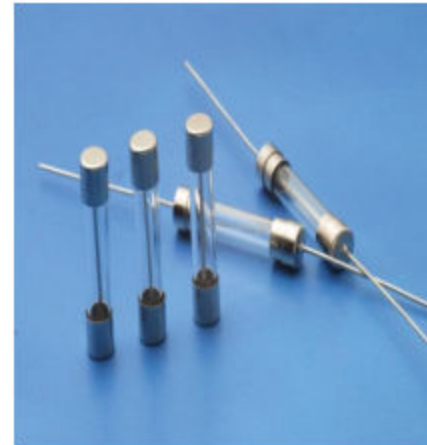
Figure 8.5: A fuse

There are different kinds of fuses but here we are introducing the most common types of fuses which we can easily find in our homes and schools. A fuse can also be called a **circuit breaker**. Have you seen these type of fuses? Explore these fuses at your school or home with an adult.

- A Cartridge fuse



Figure 8.6: Various cartridge fuse



- A Rewireable fuse



Figure 8.6: Various rewireable fuse



Static electricity and lightning

Activity 4: Understanding production of static electric charges in a balloon.

Understand how
static electric
charges can be
produced in
some common
materials

You need: A balloon and a woollen jacket.

Blow air into a balloon. Tie a knot at the end of the balloon. Now start rubbing it gently against your woollen jacket. After rubbing it for at least 30 seconds, stand in front of a mirror and hold the balloon above your head.

- What happens?
- Why does it happen?



Activity 5: Understanding production of static electric charges in a comb

You need:

A plastic comb and tiny bits of torn paper.

Take a piece of used paper and tear it into tiny bits.

Take a plastic comb and run it through your dry, non-oily hair 5 to 6 times.

Now hold the comb above the bits of paper.

What happens?

You will see that the bits of paper get instantly attracted to the comb. Why does it happen?



We have learnt earlier that atoms are made up of neutral particles called neutrons and charged particles called protons, and electrons. Electrons revolve outside the nucleus.

We have now observed that rubbing objects together make them become charged.

It means that the comb and balloon has electrical charges.

These charges are build-up on the surface of an object. They Stay around the side that was rubbed because the charges remain in

one area for a while rather than moving or "flowing" to another area, this is called static (non-moving) electricity. All materials that get charged by rubbing, and which keep their charge are said to have static electricity. Static charge is formed when two surfaces touch each other and the electrons move from one object to another. One object will have a positive charge and the other a negative charge. Rubbing the materials quickly, like when you rub a balloon fast over a woolen coat or your rub your feet on the carpet, will build up a large charge. Materials with different charges (positive and negative) will attract; while materials with similar charges (positive and positive) will push away from each other.



Figure 8.8: Creating static charge

Lightning

 Describe how lightning occurs

Have you ever seen lightning?

Have you wonder how it occurs?

Lightning is caused by clouds having charges. It occurs when oppositely charged clouds interact with each other.

Huge amount of static electric charges are build-up in clouds. When it rains we can see flashes of light, it happens because of the interaction of the built up opposite charges in clouds.



Figure 8.9: Lighting

Magnets and magnetic fields:

 Investigate the magnetic field of a bar magnet.

You have studied earlier that a **magnet** is a piece of material that attracts magnetic materials made of iron, cobalt and nickel.



Figure 8.10: A bar magnet

Have you ever seen this magnet?

What does it do?

The magnet shown above is called a bar magnet. What does 'N' and 'S' mean?

Activity 6: Understanding poles of a bar magnet

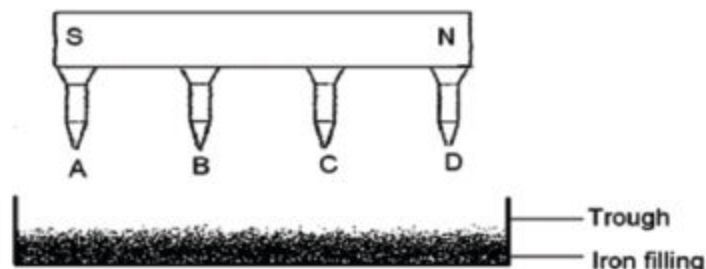
You need a bar magnet and an iron nail.

1. Place a bar magnet.
2. Bring an iron nail near it; first near the end N, then near the end S, and then near the middle of the bar magnet.
3. What do you observe? At which point does the magnet attract the nail more?

You have studied that the area around a magnet in which it attracts another object is called a magnetic field. This field is strongest at the poles. The strength of this field becomes weak as we move away from the magnet.

Activity 7: Understanding the strength of magnetic field of a bar magnet

You need a bar magnet, four small iron nails, a trough having iron fillings
Four iron nails labelled A, B, C and D, which are attached to a bar magnet as shown:



- Predict, which nail is (or nails are) able to attract the largest amount of iron filings?
- What does this prove?

Activity 8: Understanding magnetic field of a bar magnet

You need a bar magnet, a sheet of thin cardboard, iron fillings and a pencil.

- Place the bar-magnet under the cardboard.
- Sprinkle the iron fillings over the cardboard.
- Then tap the card board gently at the corner.

Observe what happens. You will observe that the iron fillings line up along the magnetic field lines. Now you can sketch the field lines using a pencil.

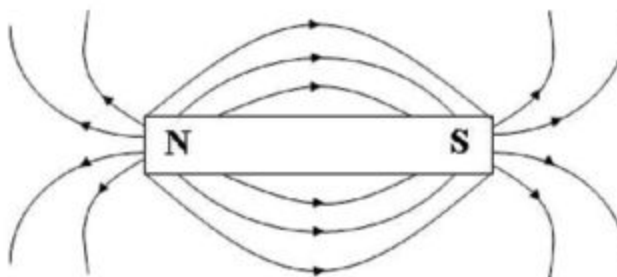


Figure 8.11: The magnetic field of a bar magnet

Activity questions:

- Where on the cardboard are iron fillings most concentrated? Why?
- What is the direction of the magnetic lines of force?

Electromagnet:

An **electromagnet** is a magnet that operates on electricity. An electromagnet is a type of magnet in which the magnetic field is induced by the flow of an electric current through a coil of wire. The magnetic field disappears when the flow of electricity is stopped.

Activity 9: Building a simple electromagnet

- 1. Construct an electromagnet and describe its working.
- 2. Explain the association between magnetism and electricity in various electromagnetic devices.

You need an iron rod or a two-inch iron nail, a thin insulated copper wire, a battery, few small iron nails or common pins, a sheet of thin cardboard, iron fillings and a pencil.

- Spin the copper wire around the rod as shown in the figure.
- Connect the ends of the copper wire to the battery.
- Bring some iron nails near one end of the rod.

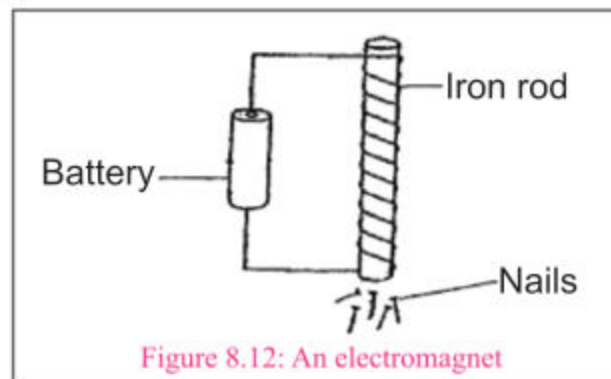


Figure 8.12: An electromagnet

What do you observe?

- Identify electromagnetic devices of everyday use.
- Explain the association between magnetism and electricity in various electromagnetic devices.

This type of magnet is called an electromagnet. It is the type of magnet which acts as a magnet only when an electric current flows through it.

Electromagnetic devices:

There are many devices which have electromagnets.

These include electric bell, electromagnetic cranes and electric motor.

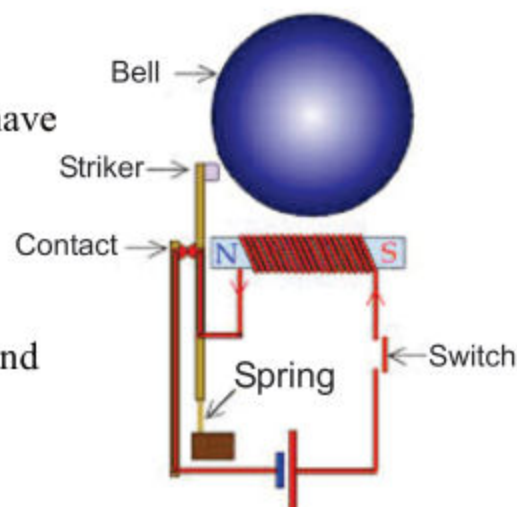


Figure 8.13: An electric bell

The electric bell:

When the current flows through the circuit, the electromagnet attracts the springy metal arm. The arm hits the gong, which makes a sound.

The electric motor:

A simple electric motor has a coil of wire. This coil can rotate between two opposite magnetic poles. When an electric current flows through the coil, the coil starts to rotate.

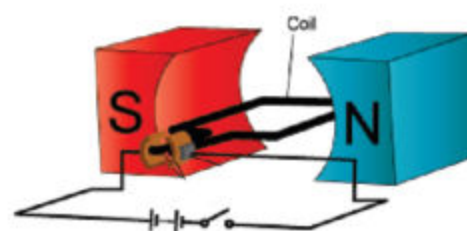


Figure 8.14: An electric motor

Do you know? A maglev (magnetically levitated) train doesn't use a regular engine like a normal train. Instead, electromagnets in the track produce a magnetic force that pushes the train from behind and pulls it from the front.



Earth's magnetic field:

T Describe the Magnetic field of the Earth and connect this concept with the use of a magnetic compass.

Have you ever seen a compass needle?

No matter where you turn it, the

needle of a compass will always points

to the North and South.

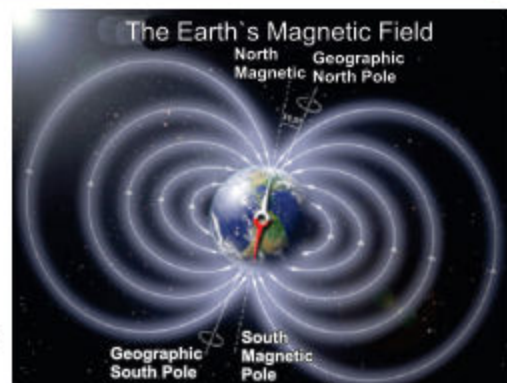


Figure 8.15: The Earth's magnetic field

It is because the Earth is like a giant magnet. It has a North pole and a South pole. This giant magnet is what makes a compass work.

Do you know? The Earth's magnetic field is caused by hot metals circulating in the core of the planet. This circulation creates electrical currents, which in turn create a magnetic field.

Making a simple magnetic compass

T Construct a magnetic compass and describe its working.

900 years ago, the Chinese were the first, to discover “magnetite”. They found out that if a piece of magnetite is placed in a dish full of water, the piece floats in water in such a manner that it always settles in the North-South direction. This discovery became the basis for the invention of a **magnetic compass**.



Figure 8.16: A compass

A **Magnetic** compass is used as a navigational tool on land and at sea. The compass detects the Earth's magnetic field. As we know that all magnetic fields have two poles, the earth's magnetic field also has two poles.

A magnetic compass consists of a light weight magnet, a magnetized needle on a free rotating pivot. Hence, the needle is attracted to the Earth's natural magnetic pole.

Activity 10: Making a magnetic compass and show its working.

What I need:

- a sewing needle
- a plastic bottle top or cork
- a pair of scissors
- a glue
- a bar magnet
- a shallow dish of water

What to do:

- Magnetize the needle by rubbing it from tail to tip with the N-pole of the bar magnet about 50 times.
- Stick the needle on the up-turned plastic bottle or cork.
- Place it carefully in the dish full of water.

Observe:

Does the needle rotate towards the N-pole?

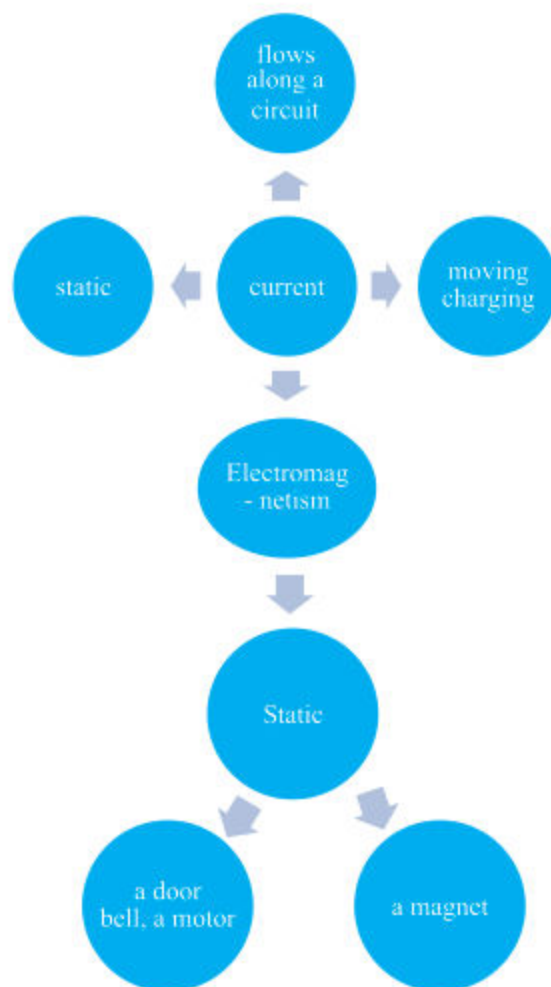
(If the tail of the needle points towards N-pole, then you are successful in your experiment)

- Now turn the cap in various directions. What is the direction of the needle?



Teacher Note: Facilitate the students in finding out the North and South directions of the Earth.

Summary



Review questions:

Tick the correct answer.

1. Which of the following is not a magnetic material?
 - a. steel
 - b. cobalt
 - c. aluminium

- 2. What happens when the two same poles of magnets are brought close to each other?**
 - a. They attract.
 - b. They repel.
 - c. They make an electric current flow.
- 3. Which of the following is the main difference between electromagnets and bar magnets?**
 - a. Bar magnets can be turned off but electromagnets cannot.
 - b. Bar magnets have a magnetic field but electromagnets do not have them.
 - c. Electromagnets need electricity but bar magnets do not need electricity.
- 4. A child rubbed two balloons with a piece of wool. Which of the following happens when the balloons are brought near each other?**
 - a. The balloons will repel each other
 - b. The balloons will attract each other
 - c. The balloons will pop
- 5. Which of the following performs its function by converting electrical energy into motion?**
 - a. an electric bulb.
 - b. an electric motor.
 - c. an electric iron.
- 6. Which of the following is the main characteristic of electromagnets that makes them useful in cranes and door bells?**
 - a. They are very strong.
 - b. They cannot transmit electricity.
 - c. They can be switched on and off.
- 7. You can sprinkle iron filings around a magnet. Where will the iron filings be attracted more?**
 - a. at poles
 - b. a way from the magnet
 - c. towards the centre of the magnet

8. Write T for True and F for False for each of the following statements.

- a) A static current flows in the wires of our home. T/F
- b) The flow of current can be increased by increasing the number of turns around the iron nail. T/F
- c) When a glass rod is rubbed by a silk cloth, it becomes charged. T/F
- d) Steel clips are not attracted by a magnet. T/F
- e) Static charges cannot flow. T/F

9. Explain why a freely suspended magnet rests in a North-South direction.

- a) What is the magnetic field of a magnet?
- b) How can the magnetic field lines of a bar magnet be observed?
- c) How does an electric motor work?
- d) What is the relationship between magnets and electricity?

Project

Materials needed:

- A small toy car
- 3 bar magnets

Procedure:

1. Tape a bar magnet to a small toy car with the North pole at the back of the car and the South pole at the front.
2. Put the car on a hard surface, like a floor or a table. Hold the bar magnet behind the car with the North pole facing the car. As you move it near the car, what happens? The North pole of your magnet repels the North pole of the magnet on the car, making the car move forward.
3. Have someone else hold another magnet in front of the car, with the North pole facing the car. Does the car move faster with one magnet "pushing" from behind and the other magnet "pulling" from ahead?
4. You can take one more car and tape another bar magnet to it. Have a car race with your friends and enjoy.